

ELEMENT A

PRESENTATION AND JUSTIFICATION OF PROBLEM

Problem Statement

28,000 American homeowners across the country have reported recurring neck, shoulder, and back pain from picking up and disposing of leaves in their yards. This has led to leaves being left in yards, causing leaf smothering and preventing grass from receiving sufficient sunlight, hindering growth. Overall, leaf removal is a task that can cause physical strain when done and, when neglected, affects the yard's health, leading the grass to die off during the fall.

Supporting Facts and Articles

In the United States, leaf removal remains a seasonal issue that requires greater attention due to the volume of leaves that fall each year. To observe, the District of Columbia collected between 8,000 and 10,000 tons of leaves between November and January. On a smaller scale, it was reported that in one acre of forest, two tons of leaves can be piled.

A National Wildlife Federation survey found that 52 percent of people are required to remove their Autumn leaves by either a homeowner's association, city, or municipality ordinance, or a combination of both. Yet, 15% of respondents decide to leave the leaves in their yard, potentially facing fines or other repercussions.¹

While not a problem year-round, leaf removal has been labeled as a tedious task in articles that surveyed homeowners on what they do with their leaves: 21 percent of those surveyed were worried about leaf smothering, and 14 percent had to dispose of around ten bags of leaves every year, and an additional six percent disposed of four to six bags per year.²

¹ Miezio, Adam. "Get Rid Of Injuries Raking Leaves Once And For All", *Fringe Sport*, 09 October 2018
https://www.fringesport.com/blogs/news/get-rid-of-injuries-raking-leaves-once-and-for-all?srsltid=AfmBOoo_ElFKx-Je-PkZOMYf-nVkg6KwJZOKWD6g0tHG8zhBpAB-rAOg

² Fallon, Cait. "To Leave (the Leaves) or not to Leave", *National Wildlife Federation*, 02 October 2023
<https://www.nwff.org/Latest-News/Press-Releases/2023/10-2-23-Leave-the-Leaves>



Figure A.1: Diagram displaying proper posture and technique for raking ³

Figure A.1 shows proper posture and technique for raking and picking up leaves. Failing to follow these guidelines, such as holding the rake close to the body and using one hand on top for leverage, can increase the risk of strain or injury, including back, shoulder, or knee pain.

³ SPT Staff. "Staying Safe With Fall Yardwork", *Somersworth Physical Therapy*, 14, October 2023
<https://somersworthpt.com/2023/10/14/staying-safe-with-fall-yardwork/>

Market Research

A. Leaf Remover Market Size

Currently, millions of American homeowners must manage the buildup of leaves in their yards, a routine task that often leads to neck, shoulder, and back strain from repeatedly bending, raking, lifting, and bagging debris. When these leaves are left on the ground, they can cause leaf smothering, blocking sunlight to the grass underneath and cutting air circulation, hindering grass growth.

The broader lawn and garden equipment market, valued at over \$31.52 billion in 2021, is expected to reach \$50.85 billion in 2028.⁴ As a result, demand for safer, more efficient leaf-removal solutions continues to grow, indicating significant potential for profit in this market.

B. Market Competitors and Barriers to Entry

Some notable products to combat this problem already on the market include: The Gardena Leaf and Lawn Collector (Figure A.2), the Ryobi 40V VAC Attack Cordless Battery Leaf Vacuum (Figure A.3), and the Worx LeafPro Universal Leaf Collection System (Figure A.4) that connects to an already existing Leaf Blower/Vacuum. These solutions address the issue of spending significant time cleaning a yard, but they all require manual labour. There is currently no autonomous solution.



Figure A.2: The Gardena Leaf and Lawn Collector⁵

⁴ Lawn & Garden Equipment Market Size, Share & COVID-19 Impact Analysis, By Equipment Type <https://www.fortunebusinessinsights.com/lawn-garden-equipment-market-106385>

⁵ “Lawn and Leaf Collector.” Gardena, 2025, us.gardena.com/products/lawn-and-leaf-collector.



Figure A.3: Ryobi 40V VAC Attack Cordless Battery Leaf Vacuum⁶



Figure A.4: Worx LeafPro Universal Leaf Collection System⁷

Figures A.2, A.3, and A.4 show solutions for removing leaves. Manufacturers such as Ryobi have created a Leaf Vacuum, which provides an easier but more expensive alternative to a rake. Another solution offered by Gardena is a Lawn Sweeper. While being a simple alternative, it requires more effort than the Leaf Vacuum and is still very expensive. WORX created a much cheaper alternative that attaches to an already existing vacuum, but still requires intense manual labour. A solution needs to be cost-effective and require minimal effort for the user.

⁶ “Ryobi 40V VAC Attack Cordless Battery Leaf Vacuum” <https://www.ryobitools.com/products/46396022899?>

⁷ “WORX LeafPro Universal Collection System Black WA4054.2 - Best Buy.” Bestbuy.com, 2020, www.bestbuy.com/product/worx-leafpro-universal-collection-system-black.

C. Consumers within the Leaf Removal Market

Consumer research shows that maintaining a clean, healthy lawn remains a strong priority for households across the country. Products that reduce physical strain and simplify yard maintenance are appealing to homeowners and families with large properties.

According to a 2018 National Association of Home Builders study⁸, the demographic profile of typical homeowners aligns strongly with regular yard-care responsibilities:

37% (3.3 million) of all recent home buyers were first-time buyers.

The typical home buyer is 40 years old.

The median household income of recent buyers was \$91,600.

27% of recent home buyers were racial or ethnic minorities.

37% (3.3 million) of recent buyers had children under age 18 living in the home.

These groups represent homeowners who consistently perform yard maintenance and would benefit from a leaf-removal solution that minimizes physical effort while preventing environmental damage from leaf smothering.

⁸ "Remodeling Market Poised for Growth as the Age of Owner-Occupied Homes Increases." Nabh.org, May 2025, www.nahb.org/news-and-economics/press-releases/2025/05/remodeling-market-poised-for-growth-as-the-age-of-owner-occupied-homes-increases.

Survey Data:

The current survey has 127 respondents. The survey aims to gather information about the scope of the problem and potential solutions.

Approximately how many times per season do you remove leaves from your yard in the fall?

127 responses

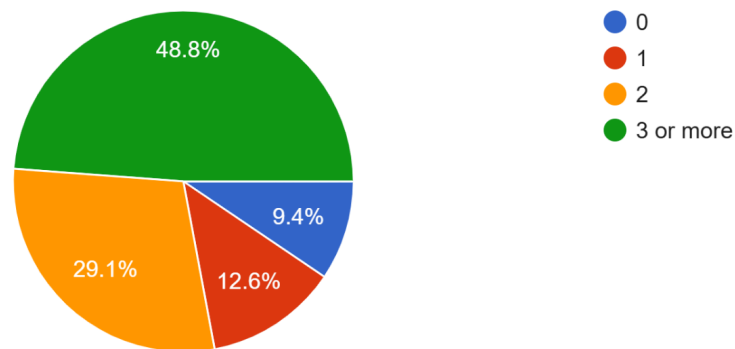


Figure A.5: Survey data for question “Approximately how many times per season do you remove leaves from your yard”

The results in Figure A.5 show that 48.8% of respondents remove leaves three or more times per season, while only 22% do so once or not at all, indicating that leaf accumulation is a frequent and recurring issue for most.

What is the total time it takes for you to remove all the leaves and dispose of them?

127 responses

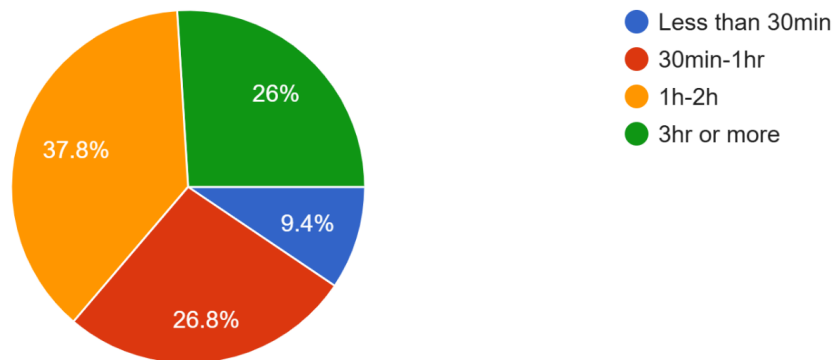


Figure A.6: Survey data for question “What is the total time it takes for you to remove all the leaves and dispose of them?”

The results of Figure A.6 show that 26% spend 3 hours or more, 37.8% spend 1–2 hours, and 26.8% spend 30 minutes to 1 hour on removing and disposing of leaves. This indicates that, for most respondents, leaf removal is a time-intensive task, taking up a large portion of their day.

Please choose any physical difficulties you experience while removing leaves. (check all that apply)

127 responses

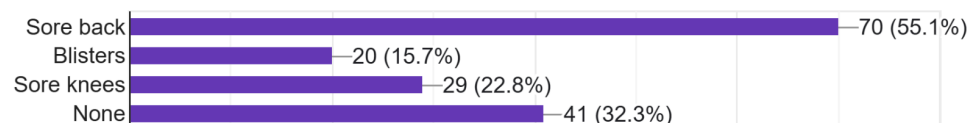


Figure A.7: Survey data for question “Please choose any physical difficulties you experience while removing leaves”

Figure A.7 shows that although about $\frac{1}{3}$ of the respondents report no physical difficulties, many experience significant difficulties, such as sore back (55.1%), blisters (15.7%), and sore knees (22.8%). These results help identify the most common issues in this problem, guiding potential solutions to address them.

If there was an alternative investment for removing up leaves, how much would you pay per cleaning?

125 responses

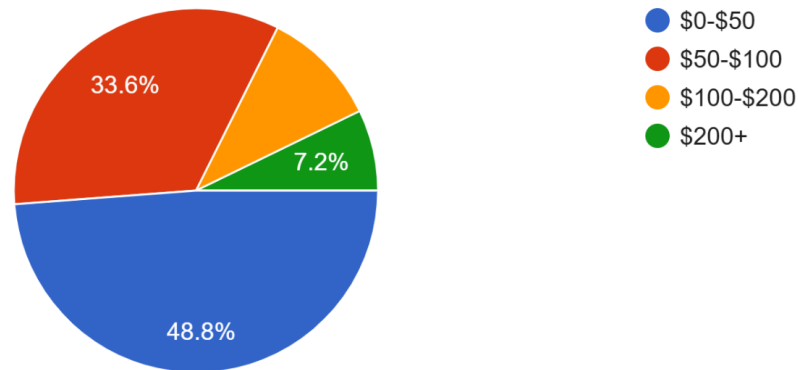


Figure A.8 : Survey data for question “If there was an alternative investment for removing leaves, how much would you pay per cleaning”

Figure A.8 shows that 48.8% of respondents would pay \$0–50 per cleaning, and another 33.6% would pay \$50–100. This suggests that most users expect a low-to-moderate cost solution, providing a clear price range to guide the solution’s complexity and feasibility.

Expert Stakeholders:

- Homeowners who regularly maintain their yards
- Environmental and conservation organizations

The primary target market is homeowners, as they are most directly impacted by the physical demands and time required for leaf removal. A solution for these problems would be highly appealing to individuals responsible for yard maintenance. Environmental organizations that focus on yard care may adopt or promote tools and systems that encourage consumers to better care for their yards and prevent issues such as leaf smothering and poor lawn health.

Conclusion:

Given the extensive research on seasonal issues, leaf removal has become a concern for people's health. Homeowners in different areas were surveyed and showed similar commonalities regarding leaf removal: physical injury and the amount of work required. Homeowners were also asked about investing in other solutions, and some indicated they needed a solution to this problem. Not every area has the same number of leaves that fall, but a solution should be developed to meet the different requirements of different regions.